

AUGER SCREW MOTOR DERIVATION

PROBLEM STATEMENT

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How do I determine motor
torque requirements to
dispense m grams of powder
in t seconds?

MOTOR RPM CALCULATION

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1. Determine the **volume per revolution** of an auger screw.

$$V_{rev} = \pi \left(\frac{D^2}{4} \right) * P$$

2. Determine **mass of powder displaced per revolution**

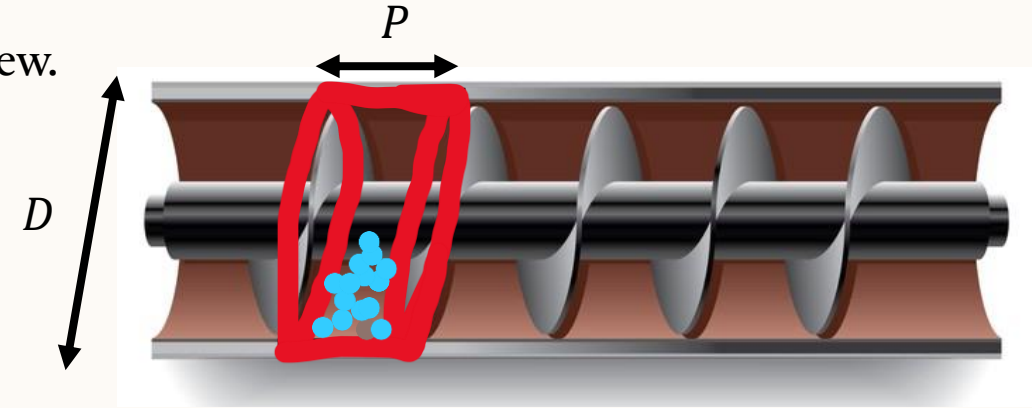
$$m_{rev} = \rho * V_{rev}$$

3. Calculate **RPM requirements** for motor

$$N = \frac{m_{dispense} * 60}{m_{rev} * t}$$

$$N_{fill} = \frac{N}{C_f}$$

Note: This is an approximate formula used to calculate speed requirements for motor! This does not take into account length of the screw!



Constants and Variables

D = Auger Screw Diameter (m)

P = Auger Screw Pitch (m)

ρ = Bulk Density of Powder ($\frac{kg}{m^3}$)

C_f = Fill Fraction Coefficient (LOOK AT NEXT SLIDE)

$m_{dispense}$ = Total Mass Dispensed (kg)

N = RPM for Motor ($\frac{rev}{min}$)

N_{fill} = RPM for Motor with Filling Factor Included ($\frac{rev}{min}$)

V_{rev} = Volume of Powder Carried per Revolution ($\frac{m^3}{rev}$)

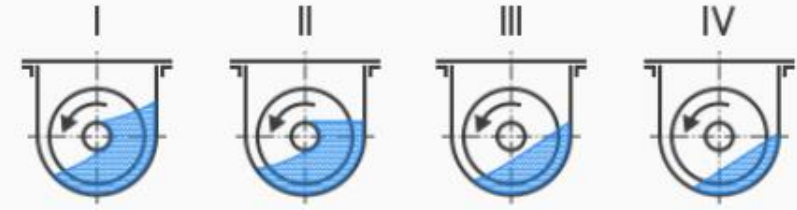
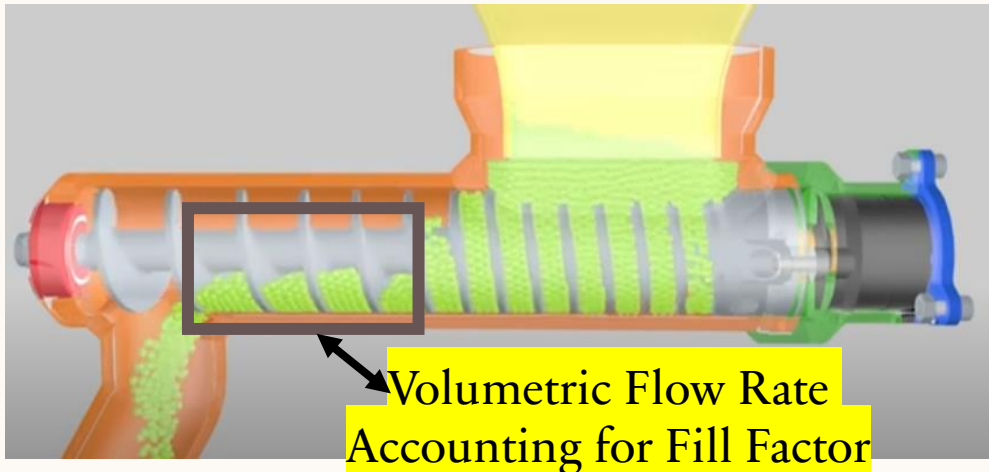
m_{rev} = Mass of Powder Carried Per Revolution ($\frac{kg}{rev}$)

TORQUE CALCULATION

1. Determine the **Screw Capacity/Volumetric Flow Rate of Powder for one revolution in the Auger**

$$Q = \frac{A * P * N}{C_f * 60}$$

Note: $C_f = 0.3 \rightarrow$ assign conservative fill factor in the range of 0.3 – 0.45 providing extra safeguard in design; protein exhibits similar flow properties to cereal flour, oats, and barley with fill factor of 0.4



I	II	III	IV
0.50	0.35	0.20	0.10

Values for individual materials are listed in the **Materials** section.

Type equation here.

Constants and Variables

D = Auger Screw Diameter (m)

P = Auger Screw Pitch (m)

ρ = Bulk Density of Powder ($\frac{kg}{m^3}$)

N = Revolutions per Minute for Motor ($\frac{rev}{min}$)

Q = Screw Capacity/Volumetric Flow Rate ($\frac{m^3}{s}$)

C_f = Fill Fraction Coefficient

A = Cross sectional area of auger screw (m^2)

FOS = Factor of Safety

TORQUE CALCULATIONS

- Determine mass flow rate

$$\dot{m} = \rho * Q$$

- Determine forces acting on FBD

$$F_{\text{powder, horizontal}} = \mu * \dot{m} * 9.81$$

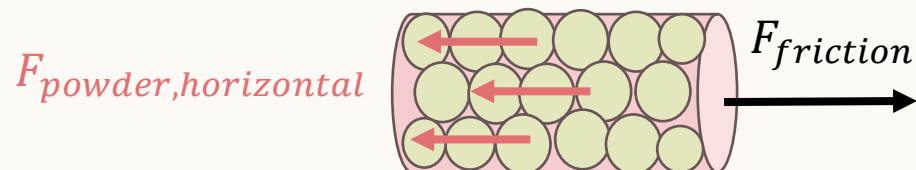
$$F_{\text{friction}} = f * V_{\text{material}} * \rho * C_f$$

$$V_{\text{material}} = A * L$$

- Find motor torque requirements with FOS

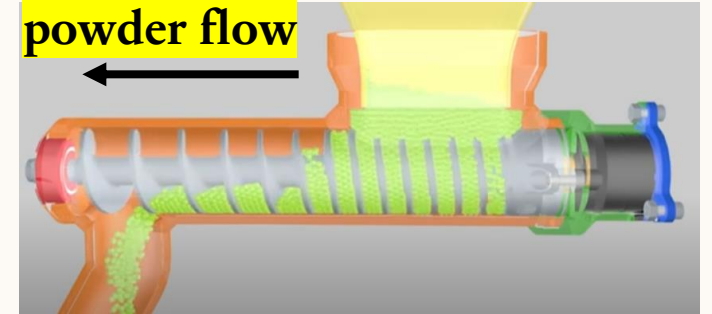
$$T_{\text{motor}} = (F_{\text{powder, horizontal}} + F_{\text{friction}}) \left(\frac{D}{2} \right) * FOS$$

In torque calculation, forces are both added together as both forces work against motor. Thus, you consider the magnitude of both forces



Note: f and μ were assigned values of **0.5** as a conservative estimate between PLA and protein powder

Direction of powder flow



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Constants and Variables

D = Auger Screw Diameter (m)

A = Cross Sectional Area of Screw (m^2)

L = Length of Auger Screw (m)

ρ = Bulk Density of Powder ($\frac{kg}{m^3}$)

$\dot{m}$ = Mass Flow Rate of Powder ($\frac{kg}{s}$)

V_{material} = Volume of Powder in Screw (m^3)

$F_{\text{powder, horizontal}}$ = Horizontal powder force against screw (N)

F_{friction} = Force to overcome friction from screw (N)

μ = coefficient of sliding friction between powder and surface

m_{dispense} = Total Mass Dispensed (kg)

f = friction coefficient between conveyor wall and powder

T_{motor} = Required motor torque specifications after FOS

FOS = Factor of Safety